Math Inquirers

Extension Workbook

with answers

Level 3

Self-study | Ages 8-10



My name _____

About this book

Math Inquirers Extension Workbook, Level 3
Self-study practice with answers.

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- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

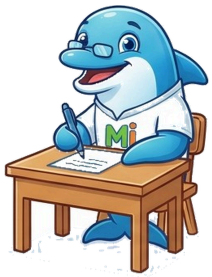
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Welcome!



This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Printing this soft copy? Print at 100 percent so the rulers are true size.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	Number sleuths (numbers to 10 000)	Not in this proof edition				☐
2	Cryptarithm crackers	Not in this proof edition				☐
3	Times table treasure	Not in this proof edition				☐
4	Model makers (bar models)	Not in this proof edition				☐
5	Remainder riddles	Not in this proof edition				☐
6	Fraction walls	6-7	8-9	10 and 12	11	☐
7	Area architects	Not in this proof edition				☐
8	Angle and line detectives	Not in this proof edition				☐
9	Measure and money masters	Not in this proof edition				☐
10	Puzzle day	Not in this proof edition				☐
11	Catch-up week	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

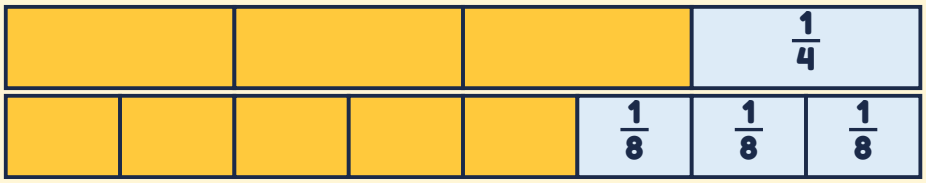
Learn it: Fractions

My goal. I can use a fraction wall to find equal fractions. I can compare fractions and add and subtract them.

A fraction wall shows one whole cut into equal parts. A longer piece of the wall is a greater fraction. To compare fractions, look at where the pieces end.

Worked example 1

Which is greater, 5/8 or 3/4? Use the fraction wall.

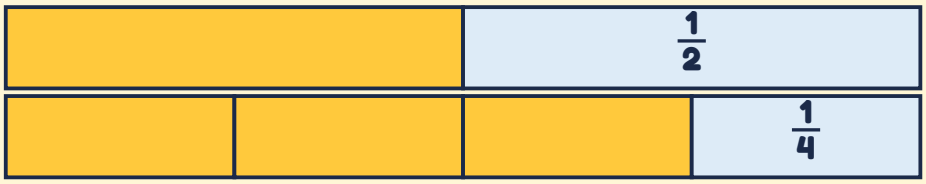


1. Shade 5/8 on the row of eighths.
2. Shade 3/4 on the row of quarters.
3. The end of 3/4 lines up with the end of 6/8.
4. 6/8 is more than 5/8, so 3/4 is greater.

Answer: 3/4 is greater than 5/8.

Worked example 2

What is 1/2 + 1/4? Use the fraction wall.



1. On the wall, 1/2 is the same length as 2/4.
2. Add the quarters: 2/4 + 1/4 = 3/4.

Answer: 1/2 + 1/4 = 3/4.

How to compare and add fractions

1. Make the parts the same size. Use the wall.
2. To add or subtract, change both to the same bottom number.
3. Add or subtract the top numbers. Keep the bottom number.

Your turn

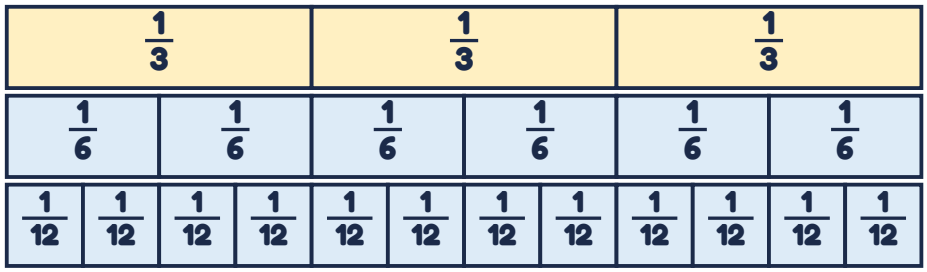
Answer at the back of the book.

Try **Circle <, = or >.** 7/12 < = > 2/3

Start here

Do these first. Take your time.

1 Use the fraction wall to fill in the blanks.

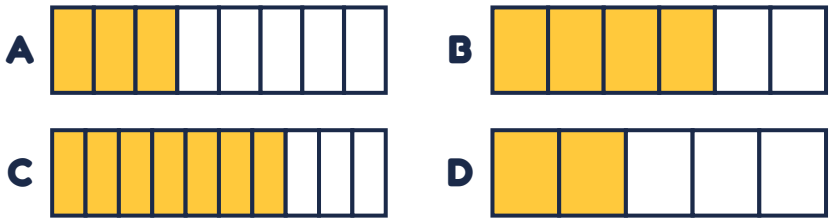


(a) 1/3 = ☐ /6 = ☐ /12

(b) 2/3 = ☐ /6 = ☐ /12

(c) 5/6 = ☐ /12

2 Write the fraction of each shape that is shaded.

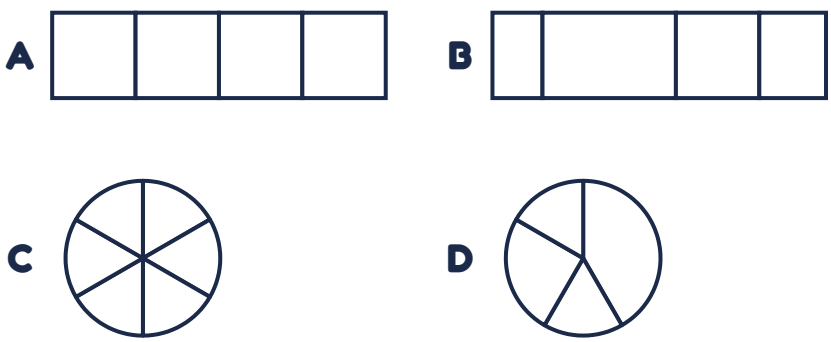


(a) A = ☐ / ☐ (b) B = ☐ / ☐

(c) C = ☐ / ☐ (d) D = ☐ / ☐

3 Which shapes are cut into EQUAL parts? Circle the letters.

A B C D



Start here

Keep going. Check your work.

4 Circle <, = or > in each part.

(a) 5/8 < = > 3/8

(b) 1/3 < = > 1/4

(c) 2/4 < = > 4/8

(d) 3/4 < = > 5/6

5 Arrange the fractions in order. Begin with the smallest.
5/6, 1/3, 1/2, 2/3

_____

6 Fill in the blanks.

(a) 3/10 + 4/10 = ☐ /10

(b) 5/8 - 2/8 = ☐ /8

(c) 9/12 - 4/12 = ☐ /12

(d) 1 - 3/10 = ☐ /10

7 Change to the same bottom number. Then fill in the blanks.

(a) 1/2 + 1/4 = ☐ /4

(b) 1/3 + 1/6 = ☐ /6

(c) 3/4 - 1/8 = ☐ /8

(d) 5/6 - 1/2 = ☐ /6

Start here

Nearly there. You can do it.

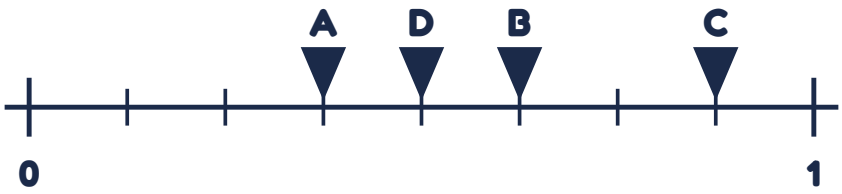
8 **A cake is cut into 12 equal slices. Ali eats 3 slices. Ben eats 4 slices.**

(a) Ali and Ben eat ☐ /12 of the cake.

(b) ☐ /12 of the cake is left.

(c) Ben eats ☐ /12 more of the cake than Ali.

9 **Look at the number line from 0 to 1. It is cut into equal steps.**



(a) A = ☐ /8

(b) B = ☐ /8

(c) C = ☐ /8

(d) Which arrow shows 1/2? A B C D

10 **Fill in the blanks to make equal fractions.**

(a) 3/4 = ☐ /12 (b) 2/5 = ☐ /10 (c) 1/2 = 5/☐

(d) 6/8 = 3/☐

11 **Siti says 3/8 + 2/8 = 5/16.**

(a) Is she right? Yes No

(b) The right answer is ☐ /8.

Think harder

Hints are upside down. Try first!

1 **Look at the fractions with bottom number 12. List every one between 1/3 and 2/3.**

_____

Working: _____

Hint. Change 1/3 and 2/3 into twelfths first.

2 **I am a fraction equal to 2/5. My bottom number is 9 more than my top number. Which fraction am I?** ☐ / ☐

My list: _____

Hint. Write fractions equal to 2/5. Look at the gap each time.

3 **A jug is full of juice. Mei drinks 1/4 of the jug. Ali drinks 1/3 of the jug.**

(a) Together they drink ☐ /12 of the jug.

(b) ☐ /12 of the jug is left.

Working: _____

Hint. Change quarters and thirds into twelfths first.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Beyond Level 3: Adding fractions with top number 1.**
Write each fraction as a sum of 2 fractions that each have top number 1.

(a) 3/4 = 1/2 + 1/ ☐ (b) 5/6 = 1/2 + 1/ ☐

(c) 7/12 = 1/3 + 1/ ☐

Hint. Take away the first fraction. What is left over?

2 **A whole bar is filled with exactly 6 pieces. The pieces are 1/2, 1/4 or 1/8 of the bar. How many of each?**

(a) ☐ halves, ☐ quarters and ☐ eighths.

My tries: _____

Hint. Eighths are the smallest. Most of the 6 pieces must be small.

3 **Colour the bar. Draw lines in 1/2 of the bar. Draw dots in 1/3 of the bar. Leave the rest white.**



(a) ☐ parts are white.

(b) That is the same as 1/ ☐ of the bar.

Hint. 1/2 of 12 parts is 6 parts. How many parts for 1/3?

4 **Make up a fraction story about a pizza cut into 8 equal slices. The answer to your story must be 5/8.**

My story: _____

My answer: _____

Hint. Pick 2 fractions with bottom number 8 that add up to 5/8.

✔ Check yourself

No hints. Try each one on your own.

1 Write the fraction that is shaded.



☐/8

2 Circle <, = or >. 5/8 < = > 3/4

Working: _____

3 5/8 − 1/4 = ☐/8

Working: _____

4 Fill in the blank. 2/3 = ☐/12

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Number sleuths (numbers to 10 000)	☐	☐	☐	☐	/ 4	
2	Cryptarithm crackers	☐	☐	☐	☐	/ 4	
3	Times table treasure	☐	☐	☐	☐	/ 4	
4	Model makers (bar models)	☐	☐	☐	☐	/ 4	
5	Remainder riddles	☐	☐	☐	☐	/ 4	
6	Fraction walls	☐	☐	☐	☐	/ 4	
7	Area architects	☐	☐	☐	☐	/ 4	
8	Angle and line detectives	☐	☐	☐	☐	/ 4	
9	Measure and money masters	☐	☐	☐	☐	/ 4	
10	Puzzle day	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

You finished the Math Inquirers Extension Workbook, Level 3.

Date

Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 6 · Fraction walls

Your turn
page 6

Answer: <
Why: 2/3 = 8/12 on the wall. 7/12 is less than 8/12, so 7/12 < 2/3.

START HERE

1
page 7

Answer: (a) 1/3 = 2/6 = 4/12 (b) 2/3 = 4/6 = 8/12 (c) 5/6 = 10/12
Why: Line up the rows. 1 third is the same length as 2 sixths and 4 twelfths. 2 thirds is 4 sixths and 8 twelfths. 5 sixths is 10 twelfths.

2
page 7

Answer: (a) 3/8 (b) 4/6 (c) 7/10 (d) 2/5
Why: The bottom number is the number of equal parts in the whole shape. The top number is the number of shaded parts. A: 3 of 8. B: 4 of 6. C: 7 of 10. D: 2 of 5.

3
page 7

Answer: A and C
Why: A fraction needs equal parts. A has 4 equal parts and C has 6 equal sectors. B and D have parts of different sizes.

4
page 8

Answer: (a) > (b) > (c) = (d) <
Why: (a) Same bottom number: 5 parts are more than 3 parts. (b) Thirds are larger pieces than quarters. (c) 2/4 = 4/8. (d) 3/4 = 9/12 and 5/6 = 10/12, and 9 is less than 10.

5
page 8

Answer: 1/3, 1/2, 2/3, 5/6
Why: Write each one in sixths: 5/6, 2/6, 3/6, 4/6. The order of the tops is 2, 3, 4, 5. So the order is 1/3, 1/2, 2/3, 5/6.

6
page 8

Answer: (a) 7 (b) 3 (c) 5 (d) 7
Why: When the bottom numbers are the same, add or subtract the top numbers and keep the bottom number. (d) 1 whole is 10/10, and 10/10 - 3/10 = 7/10.

7
page 8

Answer: (a) 3 (b) 3 (c) 5 (d) 2
Why: (a) 1/2 = 2/4, so 2/4 + 1/4 = 3/4. (b) 1/3 = 2/6, so 2/6 + 1/6 = 3/6. (c) 3/4 = 6/8, so 6/8 - 1/8 = 5/8. (d) 1/2 = 3/6, so 5/6 - 3/6 = 2/6.

8
page 9

Answer: (a) 7 (b) 5 (c) 1
Why: (a) 3/12 + 4/12 = 7/12. (b) The whole cake is 12/12, so 12/12 - 7/12 = 5/12. (c) 4/12 - 3/12 = 1/12.

9
page 9

Answer: (a) 3 (b) 5 (c) 7 (d) D
Why: There are 8 equal steps, so each step is 1/8. A is 3 steps from 0: 3/8. B is 5 steps: 5/8. C is 7 steps: 7/8. D is 4 steps: 4/8, which is half of the line.

10
page 9

Answer: (a) 9 (b) 4 (c) 10 (d) 4
Why: Multiply or divide the top and the bottom by the same number. (a) 4 × 3 = 12, so 3 × 3 = 9. (b) 5 × 2 = 10, so 2 × 2 = 4. (c) 1 × 5 = 5, so 2 × 5 = 10. (d) 6 ÷ 2 = 3, so 8 ÷ 2 = 4.

11 page 9
Answer: (a) No (b) 5
Why: When the parts are the same size, only the top numbers are added. The bottom number stays 8. 3/8 + 2/8 = 5/8.

THINK HARDER

1 page 10
Answer: 5/12, 6/12 and 7/12
Why: 1/3 = 4/12 and 2/3 = 8/12. The twelfths between 4/12 and 8/12 are 5/12, 6/12 and 7/12. The fractions 4/12 and 8/12 are not between.

2 page 10
Answer: 6/15
Why: Equal fractions of 2/5 are 4/10, 6/15, 8/20. The gap between bottom and top is 3, 6, 9, 12. The gap is 9 for 6/15. Check: 15 - 6 = 9.

3 page 10
Answer: (a) 7 (b) 5
Why: 1/4 = 3/12 and 1/3 = 4/12. (a) 3/12 + 4/12 = 7/12. (b) 12/12 - 7/12 = 5/12 of the jug is left.

GO FURTHER

1 page 11
Answer: (a) 4 (b) 3 (c) 4
Why: (a) 3/4 - 1/2 = 3/4 - 2/4 = 1/4. (b) 5/6 - 1/2 = 5/6 - 3/6 = 2/6 = 1/3. (c) 7/12 - 1/3 = 7/12 - 4/12 = 3/12 = 1/4.

2 page 11
Answer: (a) 0 halves, 2 quarters, 4 eighths
Why: Write everything in eighths: a half is 4, a quarter is 2, an eighth is 1. The 6 pieces must make 8 eighths. With no halves: 2 quarters (4) and 4 eighths (4) make 8. Using 1 half is not possible with 6 pieces: the other 5 pieces would have to make 4 eighths, but each is at least 1.

3 page 11
Answer: (a) 2 (b) 6
Why: 1/2 of the bar is 6 parts and 1/3 of the bar is 4 parts. 6 + 4 = 10 parts are coloured, so 2 parts are white. 2/12 is the same as 1/6.

4 page 11
Answer: Answers vary. Example: Ali eats 3/8 of a pizza and Mei eats 2/8 of it. Together they eat 5/8 of the pizza.
Why: Check that your fractions all have the same bottom number, and that the story gives 5/8. In the example, 3/8 + 2/8 = 5/8.

CHECK YOURSELF

1 page 12
Answer: 5
Why: The bar has 8 equal parts and 5 are shaded. The fraction is 5/8.

2 page 12
Answer: <
Why: 3/4 = 6/8. 5/8 is less than 6/8, so 5/8 < 3/4.

3 page 12
Answer: 3
Why: 1/4 = 2/8, so 5/8 - 2/8 = 3/8.

4 page 12
Answer: 8
Why: 3 × 4 = 12, so 2 × 4 = 8. 2/3 = 8/12.